Saeid Rahmani

M.Sc. Artificial Intelligence · Deep Learning for Multimodal Cancer Data

s.rahmani@ipm.ir | saeidrhm.github.io | saeidrhm | saeid-rahmani | Saeid Rahmani

Summary

Computer engineer (M.Sc., Artificial Intelligence, Sharif University of Technology) working on machine learning for cancer data. My M.Sc. thesis developed unsupervised deep-learning methods that learn structure from unlabelled neural recordings. In cancer research I analyzed TCGA gene expression and miRNA data for prognosis and diagnosis in breast cancer, lung cancer and glioblastoma, contributed to cell-free DNA studies of cancer prognosis and early detection, and built Nextflow/Cromwell pipelines for real clinical sequencing samples. I program mainly in Python and R, and I have worked with HPC systems for years, from CUDA/MPI/OpenMP research to the platform researchers at the IPM HPC Center use to run their jobs. For my PhD I want to build multimodal models that learn from longitudinal clinical, genomic, transcriptomic and residual-disease data to forecast how a patient's cancer will evolve and respond to treatment.

PhD Research Interests

- **Patient digital twins:** multimodal models that combine longitudinal clinical, genomic, transcriptomic and measurable residual disease (MRD) data for individual patients.
- **Representation learning for disease trajectories:** self-supervised and unsupervised deep learning to forecast relapse, treatment response and transformation, building on my M.Sc. work on unsupervised deep learning.
- **Liquid biopsy and molecular data:** using cell-free DNA and gene expression signals for prognosis, building on my TCGA and cfDNA work.
- **Scalable training:** training and evaluating large models on HPC and GPU clusters.

Education

M.Sc. in Computer Engineering (Artificial Intelligence) — *Tehran, Iran*

Sharif University of Technology — *2017 - 2020*

Thesis: Unsupervised neural spike sorting by deep learning methods, advisor: Dr. Ali Sharifi-Zarchi

B.Sc. in Computer Engineering — *Tehran, Iran*

Kharazmi University — *2012 - 2016*

Advisor: Dr. Azadeh Mansouri

Research and Work Experience

Pardis Gene Genetic Lab — *Tehran, Iran*

Bioinformatician — *2022 - 2025*

- Contributed to research on prognosis and early detection of cancer using cell-free DNA analysis.
- Implemented bioinformatics pipelines with Nextflow and Cromwell for analyzing real clinical sequencing samples.
- Developed computational tools for HLA typing using R and Linux-based frameworks.

Department of Genetics — *Tarbiat Modares University, Tehran, Iran*

Graduate Research Assistant — *2017 - 2022*

- Analyzed sequencing and gene expression data (TCGA datasets) using R and Linux-based tools to study prognosis and diagnosis in breast cancer, lung cancer and glioblastoma.
- Developed InterOpt, an R package and web-based tool for optimizing reference-gene selection in PCR-based gene expression normalization (published in iScience, 2023).
- Co-authored multiple cancer genetics and molecular biology studies.

High-Performance Computing (HPC) Center — *Institute for Research in Fundamental Sciences (IPM), Tehran, Iran*

Full Stack Developer — *2022 - Present*

- Lead development and maintenance of a web-based platform (Java Spring, React) for researchers to rent computational resources on the HPC cluster.
- Integrate file transfer, web-based Bash shell access and job queue management by customizing open-source tools.
- Support the functionality and security of the HPC system using extensive Linux expertise.

High-Performance Computing (HPC) Center — *Institute for Research in Fundamental Sciences (IPM), Tehran, Iran*

Graduate Research Assistant — *2014 - 2020*

- Conducted research on high-performance computing, leveraging CUDA, MPI, and OpenMP for optimized computation.
- Explored machine learning applications in medical research.
- Led internship programs in web development, machine learning, and bioinformatics.

Neuroscience Group *Royan Institute, Tehran, Iran*

Graduate Research Assistant *2017 - 2022*

- Developed novel methods for neural signal processing and analyzed multi-electrode array (MEA) data using MATLAB and R.
- Designed and implemented a C# application for real-time communication with multi-channel MEA systems, enabling closed-loop stimulation and recording.

Key Projects

Unsupervised neural spike sorting with deep learning

M.Sc. thesis, Sharif University of Technology *2017 - 2020*

- Developed unsupervised deep-learning methods to sort neural spikes from multi-electrode array recordings, without labelled data (advisor: Dr. Ali Sharifi-Zarchi).

Lung cancer subtype discrimination: in-silico analysis with experimental validation

First author, Frontiers in Genetics 2024 *2024*

- Analyzed public cancer expression data to find miRNAs that separate lung adenocarcinoma from squamous cell carcinoma, and the candidates were then validated experimentally.

InterOpt: open-source R package and web tool (interopt.ir)

First author, iScience 2023 *2023*

- Designed a weighted aggregation of reference genes that improves gene expression quantification in qPCR experiments, and released it as a package and a public web tool.

Multi-core and many-core implementation of k-means clustering

First author, MEMOCODE 2016 *2016*

- Implemented and optimized k-means clustering for multi-core and many-core processors.

fMRI computer-aided detection (3rd), fNIRS working memory (1st), Brain-Computer Interface (3rd)

Medical data analysis competitions, NBML *2020 - 2021*

Skills

Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn; unsupervised deep learning; biomedical signal analysis (MEA, fMRI, fNIRS)
Cancer Data	TCGA gene expression and miRNA analysis, cell-free DNA, HLA typing, clinical sequencing pipelines (Nextflow, Cromwell)
HPC	CUDA, MPI, OpenMP, HPC job queues, Linux, Docker, Git
Programming	Python, R, Matlab, Bash, Java (Spring), JavaScript (React), C, C++, C#
Languages	Farsi (native), English

Publications

First-author

A comprehensive in silico analysis and experimental validation of miRNAs capable of discriminating between lung adenocarcinoma and squamous cell carcinoma

Saeid Rahmani, Zahra Javanmardifard, Hadi Bayat, Hanifeh Mirtavoos-Mahyari, Mostafa Ghanei, and Seyed Javad Mowla

Frontiers in Genetics *2024*

InterOpt: Improved gene expression quantification in qPCR experiments using weighted aggregation of reference genes

Saeid Rahmani, Adel Salimi, and Ali Sharifi-Zarchi

iScience *2023*

An efficient multi-core and many-core implementation of k-means clustering

Saeid Rahmani, Armin Ahmadzadeh, Omid Hajihassani, S. Mirhosseini, and Saeid Gorgin

ACM-IEEE International Conference on Formal Methods and Models for System Design (MEMOCODE) *2016*

Cancer research and diagnostics (co-author)

Integrative analysis of urinary microRNAs for prostate cancer detection: A proof-of-concept study

LA Samani, **Saeid Rahmani**, AH Kashi, SAM Ziaee, SJ Mowla

Translational Oncology *2026*

Non-invasive discrimination of adenocarcinoma and squamous cell carcinoma based on differential expression of miR-944 and miR-326 in sputum samples of lung cancer patients

Mohammad Yazdanpour, **Saeid Rahmani**, Hadi Bayat, Hanifeh Mirtavoos-Mahyari, Adnan Khosravi, and Seyed Javad Mowla

Human Gene 2024

CLEC19A overexpression inhibits tumor cell proliferation/migration and promotes apoptosis concomitant suppression of PI3K/AKT/NF-κB signaling pathway in glioblastoma multiforme

Fatemeh Mohajerani, Zahra Moazezi Tehrankhah, **Saeid Rahmani**, Nastaran Afsordeh, Sajad Shafiee, Mohammad Hossein Pourgholami, Bahram M. Soltani, and Majid Sadeghizadeh

BMC Cancer 2024

Synthetic miR-21 decoy circularized by tRNA splicing mechanism inhibited tumorigenesis in glioblastoma in vitro and in vivo models

Hadi Bayat, Mohammad Hossein Pourgholami, **Saeid Rahmani**, Safura Pournajaf, and Seyed Javad Mowla

Molecular Therapy-Nucleic Acids 2023

Dysregulation of miR-577, miR-505-3p, miR-3682-3p, and miR-4661 in Breast Cancer Patients Based on Estrogen Receptor Status

Arman Moradi, **Saeid Rahmani**, Narges Jafarbeik Iravani, Rezvan Esmaeili, Seyed Javad Mowla, and Keivan Majidzadeh-A

Galen Medical Journal 2023

miR-361-5p as a promising qRT-PCR internal control for tumor and normal breast tissues

Sogol Ghanbari, Adel Salimi, **Saeid Rahmani**, Nahid Nafissi, Ali Sharifi-Zarchi, and Seyed Javad Mowla

Plos One 2021

Expression and function of C1orf132 long-noncoding RNA in breast cancer cell lines and tissues

Afsaneh Malekzadeh Shafaroudi, Ali Sharifi-Zarchi, **Saeid Rahmani**, Nahid Nafissi, Seyed Javad Mowla, Andrea Lauria, Salvatore Oliviero, and Maryam M. Matin

International Journal of Molecular Sciences 2021

Neuroscience, immunology and other (co-author)

Autologous Neural Stem Cell Encapsulation in CT04-Integrated Fibrin Hydrogel Enhances Tissue Regeneration After Spinal Cord Injury in Monkeys

R Jaberi, PD Okoro, M Hajinasrollah, MZ Kheirabadi, S Mirsadeghi, **S Rahmani**, Z Nazemi, I Noshadi, F Kobeissy

Molecular Neurobiology 2026

The role of the transcription factor Bcl11b in IL-17, IL-23 and/or TNF-induced activation of the canonical and non-canonical NF-kappaB pathways in psoriatic keratinocytes

S Parsa, **S Rahmani**, N Davelaar, A Mus, C Beerens, E Lubberts

British Journal of Dermatology 2024

AB0398 THE EFFECT OF INFLAMMATORY CONDITIONS ON THE REGULATION OF CANONICAL AND NON-CANONICAL NF-kB PATHWAYS VIA bcl11b

S Parsa, **S Rahmani**, N Davelaar, AMC Mus, C Beerens, E Lubberts

Annals of the Rheumatic Diseases 2024

Enhanced development of human pluripotent stem cell-derived cerebral organoids via an electrical stimulation bioreactor

Mostafa Najar-Asl, Majid Halvaei, Rouhollah Abolhasani, Sara Mirsadeghi, Susan Simorgh, **Saeid Rahmani**, Paria Pooyan, Elham Yektadoost, Sahar Kiani, Mohammad-Reza Abolghasemi-Dehaqani, and others

Chemical Engineering Journal 2024

Transplanted autologous neural stem cells show promise in restoring motor function in monkey spinal cord injury

Razieh Jaberi, Reza Jabbari, Mostafa Hajinasrollah, Sara Mirsadeghi, **Saeid Rahmani**, Masoumeh Zarei-Khairabadi, Omidvar Rezaei, Seyed-Masoud Nabavi, and Sahar Kiani

bioRxiv 2023

Stimulation of spinal cord according to recorded theta hippocampal rhythm during rat move on treadmill

Shahin Rouhi, **Saeid Rahmani**, Faezeh Shanesazzadeh, Tala Ahmadvand, Mahrokh Namazi, Mehdi Fardmanesh, and Sahar Kiani

Biomedical Engineering/Biomedizinische Technik 2023

Oral administration of lithium chloride ameliorates spinal cord injury-induced hyperalgesia in male rats

Golnoosh Rahimi, Sara Mirsadeghi, **Saeid Rahmani**, Amin Izadi, Zahra Ghodsi, Seyed Mohammad Ghodsi, Vafa Rahimi-Movaghar, and Sahar Kiani

PharmaNutrition 2022

Dopamine receptors gene expression pattern and locomotor improvement differ between female and male zebrafish during spinal cord auto repair

Parastoo Hosseini, Sara Mirsadeghi, **Saeid Rahmani**, Amin Izadi, Mohammad Rezaei, Zahra Ghodsi, Vafa Rahimi-Movaghar, and Sahar Kiani

Zebrafish *2022*

Efficient continuous skyline computation on multi-core processors based on manhattan distance

Ehsan Montahaie, Milad Ghafouri, **Saeid Rahmani**, Hanie Ghasemi, Farzad Sharif Bakhtiar, Rashid Zamanshoar, Kianoush Jafari, Mohsen Gavahi, Reza Mirzaei, Armin Ahmadzadeh, and others

2015 ACM/IEEE International Conference on Formal Methods and Models for Codesign (MEMOCODE) *2015*

Honors & Awards

Best Paper Award *Tehran, Iran*

Artificial Intelligence in Medical Imaging conference *2019*

3rd The Fifth National Competition on Functional Magnetic Resonance Imaging Data Analysis Focusing on Computer-Aided Detection (CAD) Systems *Tehran, Iran*

NBML *2020*

1st The First National fNIRS Data Analysis Competition Focusing on Working Memory *Tehran, Iran*

NBML *2021*

3rd The Fourth Iranian Brain-Computer Interface (BCI) Competition (iBCIC2021) *Tehran, Iran*

NBML *2021*

2nd *Iran*

National master entrance exam of universities in Iran *2017*

1st *Kanpur, India*

MEMOCODE design contest *2016*

2nd *Texas, USA*

MEMOCODE design contest *2015*

References

***Dr. Ali Sharifi-Zarchi*, Assistant professor** *asharifi@stanford.edu*

Computer Engineering Department, Sharif University of Technology, Iran

***Dr. Azadeh Mansouri*, Assistant professor** *a_mansouri@khu.ac.ir*

Electrical and Computer Engineering, Kharazmi University, Tehran, Iran

***Dr. Dara Rahmati*, Assistant professor** *d_rahmati@sbu.ac.ir*

Department of Computer Science and Engineering, Shahid Beheshti University, Tehran, Iran *(+98) 21 29904110*

***Dr. Seyed Javad Mowla*, Full professor** *sjmowla@modares.ac.ir*

Department of Molecular Genetics, Tarbiat Modares University, Tehran, Iran *(+98) 21 82883464*

***Dr. Sahar Kiani*, Associate professor** *sahar_kiani@royaninstitute.org*

Department of Brain and Cognitive Sciences, Royan institute, Iran *(+98) 21 23562512*